

Chapter 10

SYSTEM TESTS



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 11](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

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Section 1.0

Introduction

This chapter contains descriptions of the procedures used to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

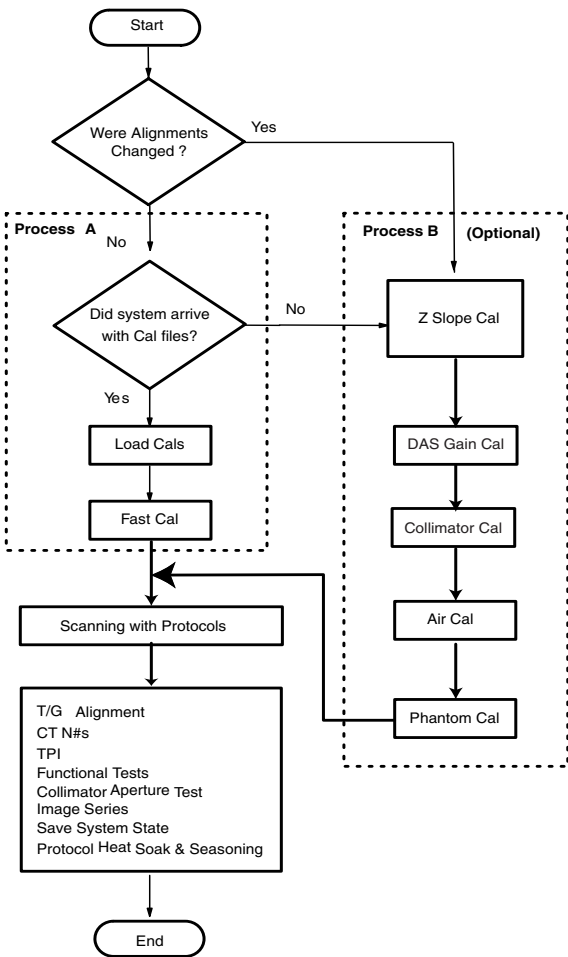


Figure 10-1 System Test Overview

Section 2.0 Calibration Process

2.1 Prepare the QA Phantom

Note: The QA phantom is shipped water filled.

- 1.) Fill the Quality Assurance (QA) phantom with distilled water, if necessary.
- 2.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 3.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.2 Calibration Process Introduction

If your system has a factory supplied state MOD, you used it to load the system calibration files during the Restore System State ([Section 5.0 on page 161](#) in Chapter 5 of this manual).

2.3 Check for the Presence of Cal Files

To determine whether or not the system contains calibration files:

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the ERROR LOGS icon.
- 3.) Select SESSION LOG.
- 4.) Select HISTORY VIEWER.
- 5.) Select CAL REPORT.
- 6.) Select VIEW to open the Cal Listing window.



- If a Cal list appears in the window, the factory supplied calibration files were loaded during the Chapter 5 procedures, you may proceed to [Section 2.4](#).

-or-

- If the system does NOT display a list of calibrations, proceed directly to [Appendix G on page 187](#).

2.4 Calibration Process A



- 1.) Select DAILY PREP to warm up the tube.
- 2.) Select FAST CAL from the Daily Prep menu.

Note: Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)

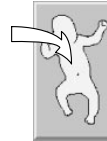
- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, continue to the Table/Gantry Alignment procedure in [Section 2.6](#).

2.5 How to Scan with Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.

- 1.) Select the NEW PATIENT icon.
- 2.) Enter the Patient ID.



- 3.) Click left on the infant anatomical figure icon:



- 4.) Click left on the area below the infant's feet (inside circle):
- 5.) Select a protocol from the list, to display the corresponding view edit screen. Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.6 Table/Gantry Alignment Procedure

- 1.) Drive the table to its highest elevation.
- 2.) Check the relationship between the cradle and the cradle drive end:
 - a.) Set the gap between the angled rollers to center the cradle over the drive end.
 - b.) Make sure the cradle does NOT touch either angled roller.
- 3.) To adjust the cradle, loosen the six cradle mounting bolts.
- 4.) Turn on the alignment lights.
- 5.) Advance the end of the cradle to the internal light.
 - The cradle end should be parallel to the light, (assuming proper alignment light function).
 - If necessary, adjust the table to bring the cradle end into parallel alignment with the light.
- 6.) Align the front edge of the cradle with the laser light.
- 7.) Select the service protocol, PERPENDICULAR ALIGNMENT.

or

Manually select the scan parameters in [Table10-1](#).

Scan Type	Direction	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm
Axial	Head First	120	40	Large	5mm/1i	2sec	S0	S0	Bone
Axial	Head First	120	40	Large	5mm/1i	2sec	I1000	I1000	Bone

Table 10-1 Perpendicular Alignment Scan Parameters

- 8.) Use over/under multi-image display to view both images at the same time.

- 9.) Orient the vertical cursor to 90 degrees, and use it to verify that the left and right edges of the cradle fall on a straight line. If the edges do NOT line up:
 - a.) Loosen the table anchors at locations #5, #6 and #7.
 - b.) Move the table half the distance.
 - c.) Return to Step 7. Repeat the procedure until the table edges line up.
 - d.) Tighten the anchors and torque to 75 6 N-m (55 5 ft.-lb.).
 - e.) Reinstall all the table parts you removed to access the anchors, including covers, ground straps, and the center support bar.
- 10.) Install the table and gantry covers.

2.7 AUTO CT'N' Number Adjustment

Note: If it has been more than 30 minutes since calibration scans, rerun FASTCAL. N Number Adjustment needs the tube to be warm, or CT numbers will change. FASTCAL takes about 15 minutes.

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Select SCANNER UTILITIES.
- 2.) Select WATER CALIBRATION.
- 3.) Follow directions on the screen.

The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

Section 3.0 System Tests

3.1 Introduction

Use the system tests in the following sections to exercise all aspects of the system, to assure system integrity, before turnover to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

3.2 Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the black line on the phantom lines up with the internal laser lights.
- 3.) Verify that BOTH internal axial lasers line up along the black line on the QA phantom. If not,

check table/gantry, cradle, and/or laser alignment.

- 4.) Center the phantom in the scan plane with the calibration program.
- 5.) Select the service protocol, TOMO PLANE INDICATION.

or

Manually select the scan parameters in [Table10-2](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

Table 10-2 Tomographic Plane Indication Scan Parameters

- 6.) Set Reconstruction for every 0.2mm
- 7.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 8.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 9.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.
- 10.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 11.) Check the box on **Form 4879**.

3.3 System Scanning Test

Use the System Scanning Test to verify hardware functionality. Review images for visible artifacts, and review the message log for unacceptable errors.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the black line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.
- Note: Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.
- or-
- Manually select the scan parameters specified in [Table10-3](#).

Scan Type	kV	mA	SFOV	Thickness	Rotation Time	Start Loc.	End Loc.	Tilt/Table Speed
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	I30
Axial (2scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	S30
Helical HQ	120	50	Large	5mm	1.0sec	S70	I75	15mm/rot
Helical HS	120	50	Large	5mm	0.8sec	S70	I75	22.5mm/rot

Table 10-3 System Scanning Test Scan Parameters

- 3.) Complete the scans.

Section 4.0 Collimator Aperture Test

4.1 Small Aperture Test

- 1.) Clear all obstructions out of the gantry SFOV.
 - 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
-
- 2.) Click on the TOOLBOX/UTILITIES icon.
 - 3.) Select INSTALL.
 - 4.) Select SYSTEM TESTS.
 - 5.) Select COLLIMATOR APERTURE TEST.
 - 6.) Select SMALL APERTURE SCAN to display a scan protocol window, and start gantry activity (refer to [Figure 10-2](#)).
 - 7.) Press the **START SCAN** key when it flashes, to initiate the first scan.
 - 8.) When the test completes, the screen updates to display the Small Aperture values.

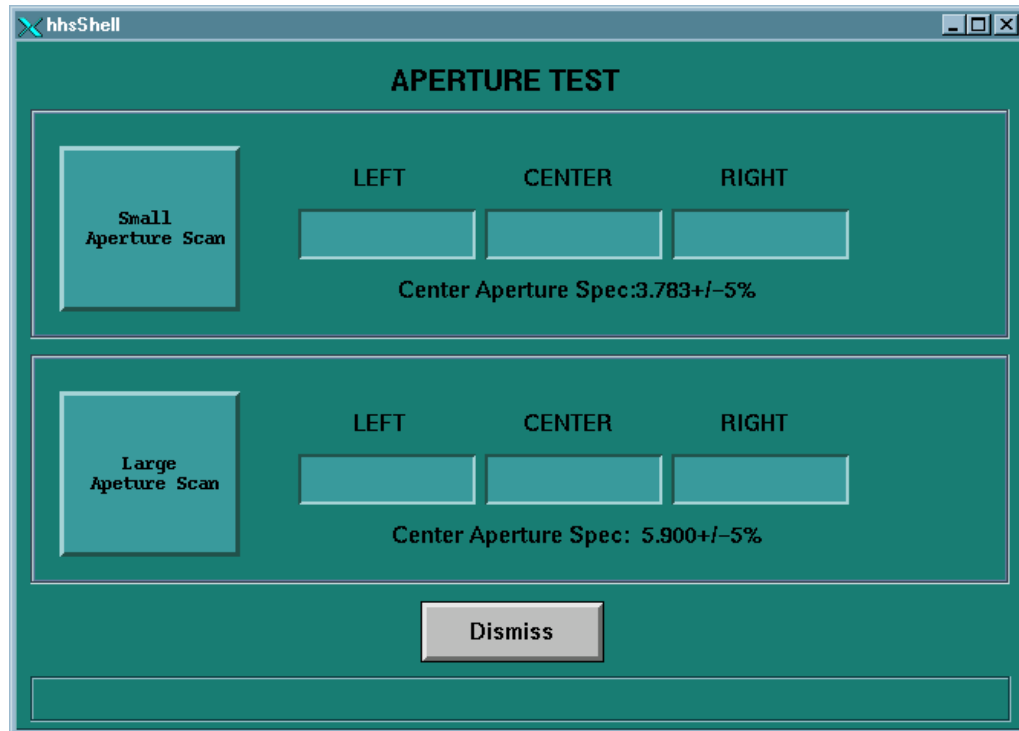


Figure 10-2 Collimator Aperture Test Screen

4.2 Large Aperture Test

Note: The center channel is the only channel with a spec. The left and right channels may or may not pass. Small aperture center spec is 3.783 +/- 5%. Large aperture center spec is 5.900 +/- 5%.

- 1.) Select LARGE APERTURE to display a scan protocol window, and start gantry activity (Refer to [Figure 10-2](#)).
- 2.) Press the START SCAN key when it flashes, to initiate the first scan.
- 3.) When the test completes, the screen updates to display the large aperture values.
- 4.) Record the large and small aperture values on Form 4879, located in [Chapter 11](#).

Section 5.0 Image Series

5.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images, and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements, and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: Consider *any* image series scan that does not meet specifications as failing.

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

5.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

5.3 Term Definitions

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

5.4 Image Series for LightSpeed QX/i System

- 1.) Place the 48cm phantom on the phantom holder.
 - Align phantom with the internal laser lights.
 - Center the phantom with the calibration program.
- 2.) Manually select the scan parameters in [Table 10-4](#) or locate a similar protocol in the Service Protocol - Image Series 48cm.

Scan Type	Phantom	kV	mA	SFOV	Thickness	Scan Time	Start/End Loc.	# of scans
Axial	48cm	120	340	Large	5mm (4i)	1.0sec	17.5/S7.5	4

Table 10-4 LightSpeed QX/i Image Series Scan Parameters

- 3.) Record the data on the data sheet, and verify the images meet specifications.
- 4.) Proceed to [Section 5.5](#), to analyze the image series.

5.5 Analyze Image Series

- 1.) Open the IMAGE WORKS desktop.
- 2.) When the Browser opens, select the first Exam/Series/Image(s) to analyze.
- 3.) Select MEASURE -> BOX ROI.
- 4.) Position the ROI box over the image in the location(s) listed in the corresponding Image Data Sheet, and record the results in each Table.
- 5.) Return to the Browser, and select the next Exam/Series/Image(s) to analyze. Repeat the process to gather data to fill all the tables on the Image Data Sheets.

5.6 QA Image Series

5.6.1 QA Image Series Procedure

- 1.) Place the QA phantom, 2144715, on the phantom holder.
 - Align with the black line with the laser lights.
 - Center with the Calibration program.
- 2.) Select the Service Protocol, IMAGE SERIES QA.
-or-
Manually select the scan parameters in [Table10-5](#).

Scan Type	Phantom Type	kV	mA	SFOV	Thick-ness	Scan Time	# of scans	Scan Length	Total images
Axial	QA#1	120	260	Small	10mm 2i	1.0sec	4	I5.0 - S5.0	8
Axial	QA#1	120	260	Small	10mm 2i	1.0sec	4	Bone Retro w/ DFOV = 15cm	8
Axial	QA#2	120	260	Small	10mm 2i	1.0sec	4	S400.0 - S50.0 Peristaltic OFF	8
Axial	QA#3	120	260	Small	10mm 2i	1.0sec	4	S55.0 - S65.0	8

Table 10-5 Image Series QA Scan Parameters

- 3.) Retrospectively reconstruct the first QA#1 scan with the bone algorithm, and 15cm Display FOV, for later analysis. Use the parameters in [Table10-5](#) to reconstruct the image.
- 4.) Record the data on the data sheet, and verify the images meet specifications.
- 5.) Proceed to [Section 5.6.2](#), to analyze the QA image series.

5.6.2 Analyze the QA Image Series

- 1.) Refer to Figure 10-5. To position the ROI box, and analyze the region of interest:
 - a.) Move: Position the mouse cursor over the box. Press and hold the left mouse button, and move the mouse to reposition the box.
 - b.) Accept: Click the right mouse button once, to accept the current position.
 - c.) Accept: Toggle the right mouse button once, to accept the remaining ROI's.
- 2.) Use the formulae displayed in Figure 10-4 to calculate the MTF and Contrast Scale, and record those results in [Table10-8](#).
- 3.) To adjust the image window (refer to [Figure 10-3](#)):

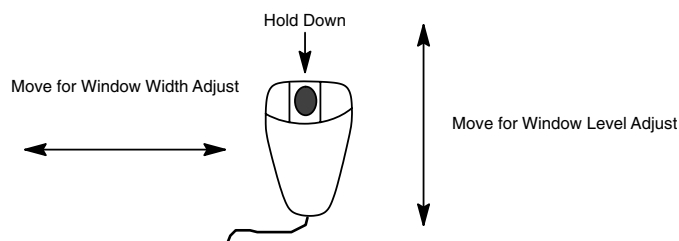
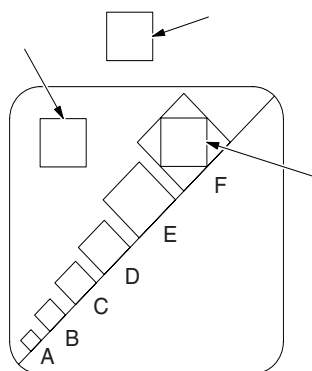


Figure 10-3 Adjust Window Width/Level with Mouse



Contrast Scale = mean of B – mean of C.

SD = SD of ROI Box over F slits.

SD = (SD + SC) / 2

Modulation = _____

MTF = 2.2 (Modulation/Contrast Scale)

Figure 10-4 MTF and Contrast Scale

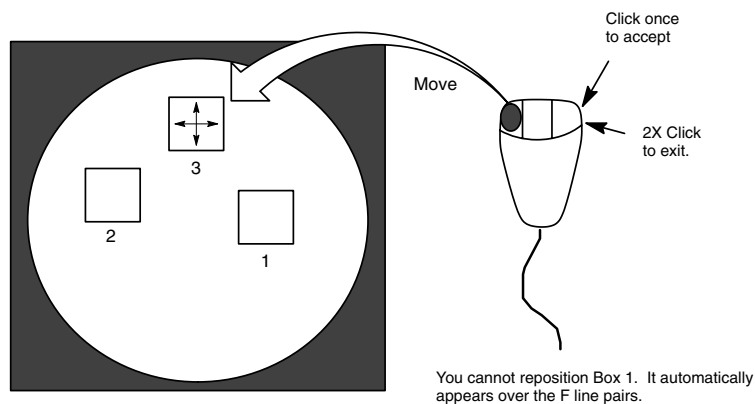


Figure 10-5 Reposition ROI with Mouse

- 4.) Record the data on the data sheet and verify the scans meet all specifications.

Section 6.0 Image Data Sheets

6.1 Image Series Data Sheet 48cm Poly Phantom

48cm/L Scan parameters: 5mm (4i)/120kV/ 340mA /1 sec./16 Images

Rows	AvXc	AvXo	AvXo - AvXc	AvSDo	Comments	Scan Tech Signature/ Date
2A						
1A						
1B						
2B						
SPECS	---	---	± 8.5	<50.0	---	---

Table 10-6 48cm Poly Phantom Image Series Data

*If your AvSDo fails, check to make sure you did not scan this technique at 1mm.

Box Size: 45 x 45 pixels

Center Coordinates: 256, 256

Outside Coordinates: 256, 60 452, 256 256, 452 60, 256

Image Acceptance/Date: _____

Certified Image Reviewer: _____

ARTIFACT LIMITS:

Rings:	Smudge:
Ring Factor: < ± 36 counts	Smudge Factor: < ± 14.0 counts
Band Radius: 0 - 23.5cm	Streaks:
Band Factor: < ± 8.0 counts	Streak Factor: < ± 4.0 counts
Clump:	
Max Clump Size: 3	

Table 10-7 Artifact Limits

6.2 Image Series Data Sheet 20cm QA#1 Phantom

QA#1 scan parameters: S/10mm (2i)/120kV/260mA/1 sec/4 images

Use this scan data for Bone Retro.

Record data on **Form 4879.1**

Exam/Series/ Image	MTF	MTF 4 slice avg.	Contrast Scale	Comments/ Artifacts	Scan Tech Signature/ Date
____/____/1		N/A			
3		N/A			
2		N/A			
4					
SPECS	- - -	0.65 to 1.0	110.0 to 130.0	- - -	- - -

Table 10-8 QA#1 Image Series Data

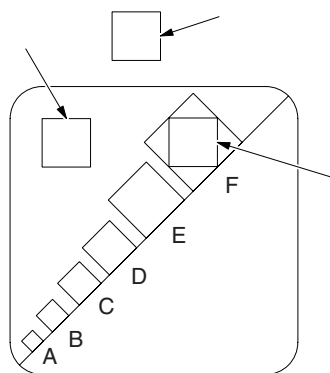
QA#1 scan parameters: S/10mm (2i)/120KV/260mA/1 sec./4 images

Bone Retro/Display FOV = 15cm

Record data on **Form 4879**

Exam/Series/ Image	Line Patterns Visible	Comments/ Artifacts	Scan Tech Signature/Date
____/____/1			
2			
3			
4			
SPECS	B, C, D, E, F	- - -	- - -

Table 10-9 QA#1 Bone Retro Data



Contrast Scale = mean of B—mean of C.

SD = SD of ROI Box over F slits.

SD = (SD + SC) /2

Modulation = _____

MTF = 2.2 (Modulation/Contrast Scale)

Figure 10-6 Box Size: 17 x 17 pixels

6.3 Image Series Data Sheet 20cm QA#2 Phantom

QA#2 scan parameters: S/4x5 (2i)/120KV/260mA/1 sec/Peristaltic OFF

Scan from S40.0 to S50.0

Record results on **Form 4879**

Exam/Series/ Image at location S40.0	Visible Hole View w/ Window 20	Contrast Factor	Comments/ Artifacts	Scan Tech Signature/Date
____/____/1				
3				
5				
7				
SPECS	See Table 10-11	2.0 to 12.0	---	---

Table 10-10 QA#2 Data Table

Box size: 49 x 9 pixels

Contrast Factor	Visible Number of Holes: Lower Limit*	Visible Number of Holes: Upper Limit*	Smallest Visible Hole: Size
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

*Required number of visible holes depends on the contrast factor.

Table 10-11 Visible Hole Table

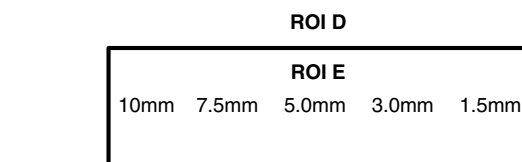
2 out of the 4 slices taken must meet this specification.

Image Acceptance/Date: _____

Certified Image Reviewer: _____

For Reference Only:

Use the manual method for calculating Contrast



ROI D 49 X 9 pixels x=256, y=200
ROI E 49 X 9 pixels x= 256, y=235
Contrast Factor = Mean E – Mean D

Factor.

6.4 Image Series Data Sheet 20cm QA#3 Phantom

METHOD:

- 1.) Use Series Means tool to obtain AvXc and AvXo - AvXc calculations.
- 2.) Use the QA#3 tool to obtain AvSDc calculations.
- 3.) QA#3 scan parameters: S/10mm 2i/120KV/260mA/1 sec./8 images
- 4.) Scan from S55.0 to S65.0

- 5.) Record data on **Form 4879**.

Rows	AvXc	AvSDc	AvXo - AvXc	Comments/Artifacts	Scan Tech Signature/Date
2A1A					
1B2B					
SPECS	± 3.0	2.6 to 3.4	± 3.0	---	---

Table 10-12 QA#3 Data

Comment: The average value of each row in [Table 10-12](#) must meet the specifications listed at the bottom of the corresponding column.

- QA#3 Box Size: 51 x 51 pixels at Coordinate 256, 256
- Series Means Box Size: 31 x 31 pixels at Coordinate 256, 256
- Center Coordinates: 256, 256
- Outside Coordinates: 256, 95
 - 417, 256
 - 256, 417
 - 95, 256

ARTIFACT LIMITS:

- **Rings:** Ring Factor: < ± 4.8 counts
- **Band:** Radius 0 to 8.5cm = Band Factor: ± 2.8 counts
- **Band:** Radius 8.5cm = Band Factor: < ± 2.8 counts
- **Center Artifact:** Center Artifact Factor: 3.5 Std. Deviation.
- **Smudge:** Smudge Factor: < ± 2.2 counts
- **Streaks:** Streak Factor: < ± 4.0 cts

Image Acceptance/Date: _____

Certified Image Reviewer: _____

Section 7.0 Save System State

Use the following commands to create the System State MOD.

- 1.) Load an MOD into the drive port on the front of the console.
- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 3.) Click on the PM icon.
- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.
- 5.) Select ALL
- 6.) Select SAVE
- 7.) When the save operation completes, select FILE and QUIT from the pull down menu.
- 8.) Remove the MOD from the drive

Section 8.0

Tube Heat Soak and Seasoning

Refer to [Section 3.0 on page 229](#) of Appendix H for the Tube Heat Soak and Seasoning process.

Section 9.0

Additional Procedures and Information

9.1 Exposure Time Accuracy

- 1.) Turn on Monitor Enable to read the scan times.
 - a.) Select END EXAM to halt any exam in progress. This step turns on the Monitor Enable function.
 - b.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
 - c.) Click on the TOOLBOX/UTILITIES icon.
 - d.) Select INSTALL.
 - e.) Select GENERATOR CALIBRATION.
 - f.) Select START to open the Generator Installation and Verification menu.
 - g.) Select BLEEDER SETUP (DDC) to open the Diagnostic Data Collection menu.
 - h.) Select MONITOR ENABLE to display the scan times in the message log. Do NOT dismiss, or close, the DDC window, because it turns OFF the monitor enable function.
- 2.) Toggle to the Exam Rx desktop.
- 3.) Select the Exposure Time Accuracy Service Protocol.

or

Manually select the scan parameters specified in [Table10-13](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Table Speed
Scout	120	40	-	-	-	S10	I10	-
Scout	120	40	-	-	-	S75	I75	-
Scout	120	40	-	-	-	S300	I300	-
Axial	120	40	Large	1.25mm 4i	1.0sec	S0		-
Axial	120	40	Large	1.25mm 4i	4.0sec	S0		-
Helical	120	30	Large	5 HQ	-	I212	S213	15mm/sec

Table 10-13 Exposure Time Accuracy Scan Parameters

- 4.) Complete the scans and record the scan times displayed in the message log, on **Form 4879**.
- 5.) When complete, toggle to the Service Desktop and DISMISS the DDC window to turn OFF the Monitor Enable function. Failure to turn OFF Monitor Enable fills the message log with kV, mA, and scan times.

9.2 Streak Test

- 1.) Place the 48cm phantom on the phantom holder.
 - Align the phantom to the internal laser lights.
 - Use the Calibration program to center the phantom.

- 2.) Select the Service Protocol, STREAK TEST.

or

Manually select the scan parameters in [Table10-14](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	# of scans
Axial	120	200	Large	5mm/4i	1.0sec	30

Table 10-14 Streak Test Scan Parameters

- 3.) Evaluate all images for streaks or rings.
- 4.) Fix all streak or ring failures. All 120 images should be ring and streak free.

9.3 Artifact Definitions

Refer to the System Service manual for an explanation of image quality artifacts.

Appendix I

OPTIONAL PROCEDURES

Section 1.0

Calibration Process B

Take care to align the water filled area of the phantom in the SFOV during water cals. Locate the black positioning markers over the resolution portion of the phantom. Scan the section of the phantom that starts about 60mm ($2\frac{3}{8}$ inches) in front of the black positioning markers.

Follow this calibration process when the system arrives from the factory without a set of calibration files.

- 1.) Perform Z-Slope Cal by selecting the function from the Service Desktop. It runs air scans. Make sure x-ray beam is clear.
- 2.) Select SCANNER UTILITIES.
- 3.) Select DETAILED CALIBRATION.
- 4.) Choose the user select option.
- 5.) Perform DAS Gain Calibration by selecting the function from Scanner Utilities.
 - Select SCANNER UTILITIES and select DAS GAIN CALIBRATION.
 - Ensure that there is nothing in the x-ray beam and continue.
 - The system will now perform a mylar window check and provide the appropriate messages if the window should need cleaning.
 - Upon completion of the mylar window scans the system will now take 31 scans and save the results in the calibration database. The system will provide the appropriate messages if the calibration should fail.
- 6.) Perform Collimator Calibration by selecting the function from Scanner Utilities.
 - Select SCANNER UTILITIES and select COLLIMATOR GAIN CALIBRATION.
 - Ensure that there is nothing in the x-ray beam and continue.
 - The system will now perform a mylar window check if needed and provide the appropriate messages if the window should need cleaning.
 - Upon completion of the mylar window scans the system will now take 8 scans and save the results in the calibration database. The system will provide the appropriate messages if the calibration should fail.
- 7.) Use a bubble level and the Z-axis knob on the phantom holder to level the phantom.
- 8.) Select the following parameters from the Air Calibration screen.
 - a.) Calibration Type: Small/Head, Large/Body
 - b.) KV: Select all kV settings.
 - c.) Aperture: All
 - d.) Focal Spot: All
- 9.) Acquire the Air and Phantom Calibrations. When the scans complete, windows open with detailed instructions to center and scan the calibration phantoms.
- 10.) When you finish the calibration process, continue to the Table/Gantry Alignment procedure, [section 2.6 on page 234](#).

Section 2.0

MANUAL CT N Number Adjustment

Note: If it has been more than 30 minutes since calibration scans, rerun FASTCAL. N Number Adjustment needs the tube to be warm or CT numbers will change. FASTCAL takes about 15 minutes.

Run the small and large phantom CT N number adjustment before you acquire the Image Series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Select EXAM RX.
- 2.) Place the QA phantom on the phantom holder. Align the water section with the laser lights. Take care to align the water filled area of the phantom in the SFOV during water cals. Locate the black positioning markers over the resolution portion of the phantom. Scan the section of the phantom that starts about 60mm (2 ³/₈ inches) in front of the black positioning markers.
- 3.) Select SCANNER UTILITIES.
- 4.) Select CENTER PHANTOM from the Scanner Utilities Menu.
- 5.) Center the Phantom.
- 6.) Select the Service Protocol CT N NUMBERS.
- 7.) Set the center of the water section of the QA phantom to S0.0.
- 8.) Use the scan protocols listed in [Tablel-1](#) for 80kV, [Tablel-2](#) for 100kV, [Tablel-3](#) for 120IV, and [Tablel-4](#) for 140kV.
- 9.) Complete all scans in the CT N numbers protocol.
- 10.) Proceed to [section Section 3.0 on page 252](#).

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	80	4x1.25	I1.8	S1.8	Small	Small	200
2	80	4x1.25	I1.8	S1.8	Large	Small	200
3	80	4x2.5	I3.8	S3.8	Small	Small	200
4	80	4x2.5	I3.8	S3.8	Large	Small	200
5	80	4x3.75	I5.6	S5.6	Small	Small	200
6	80	4x3.75	I5.6	S5.6	Large	Small	200
7	80	4x5	I7.5	S7.5	Small	Small	200
8	80	4x5	I7.5	S7.5	Large	Small	200
9	80	4x1.25	I1.8	S1.8	Small	Large	350
10	80	4x1.25	I1.8	S1.8	Large	Large	350
11	80	4x2.5	I3.8	S3.8	Small	Large	350
12	80	4x2.5	I3.8	S3.8	Large	Large	350
13	80	4x3.75	I5.6	S5.6	Small	Large	350
14	80	4x3.75	I5.6	S5.6	Large	Large	350
15	80	4x5	I7.5	S7.5	Small	Large	350
16	80	4x5	I7.5	S7.5	Large	Large	350

Table I-1 80kV Protocols

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	100	4x1.25	11.8	S1.8	Small	Small	200
2	100	4x1.25	11.8	S1.8	Large	Small	200
3	100	4x2.5	13.8	S3.8	Small	Small	200
4	100	4x2.5	13.8	S3.8	Large	Small	200
5	100	4x3.75	15.6	S5.6	Small	Small	200
6	100	4x3.75	15.6	S5.6	Large	Small	200
7	100	4x5	17.5	S7.5	Small	Small	200
8	100	4x5	17.5	S7.5	Large	Small	200
9	100	4x1.25	11.8	S1.8	Small	Large	300
10	100	4x1.25	11.8	S1.8	Large	Large	300
11	100	4x2.5	13.8	S3.8	Small	Large	300
12	100	4x2.5	13.8	S3.8	Large	Large	300
13	100	4x3.75	15.6	S5.6	Small	Large	300
14	100	4x3.75	15.6	S5.6	Large	Large	300
15	100	4x5	17.5	S7.5	Small	Large	300
16	100	4x5	17.5	S7.5	Large	Large	300

Table I-2 100kV Protocols

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	120	4x1.25	11.8	S1.8	Small	Small	200
2	120	4x1.25	11.8	S1.8	Large	Small	200
3	120	4x2.5	13.8	S3.8	Small	Small	200
4	120	4x2.5	13.8	S3.8	Large	Small	200
5	120	4x3.75	15.6	S5.6	Small	Small	200
6	120	4x3.75	15.6	S5.6	Large	Small	200
7	120	4x5	17.5	S7.5	Small	Small	200
8	120	4x5	17.5	S7.5	Large	Small	200
9	120	4x1.25	11.8	S1.8	Small	Large	250
10	120	4x1.25	11.8	S1.8	Large	Large	250
11	120	4x2.5	13.8	S3.8	Small	Large	250
12	120	4x2.5	13.8	S3.8	Large	Large	250
13	120	4x3.75	15.6	S5.6	Small	Large	250
14	120	4x3.75	15.6	S5.6	Large	Large	250
15	120	4x5	17.5	S7.5	Small	Large	250
16	120	4x5	17.5	S7.5	Large	Large	250

Table I-3 120kV Protocols

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	140	4x1.25	11.8	S1.8	Small	Small	170
2	140	4x1.25	11.8	S1.8	Large	Small	170
3	140	4x2.5	13.8	S3.8	Small	Small	170
4	140	4x2.5	13.8	S3.8	Large	Small	170
5	140	4x3.75	15.6	S5.6	Small	Small	170
6	140	4x3.75	15.6	S5.6	Large	Small	170
7	140	4x5	17.5	S7.5	Small	Small	170
8	140	4x5	17.5	S7.5	Large	Small	170
9	140	4x1.25	11.8	S1.8	Small	Large	200
10	140	4x1.25	11.8	S1.8	Large	Large	200
11	140	4x2.5	13.8	S3.8	Small	Large	200
12	140	4x2.5	13.8	S3.8	Large	Large	200
13	140	4x3.75	15.6	S5.6	Small	Large	200
14	140	4x3.75	15.6	S5.6	Large	Large	200
15	140	4x5	17.5	S7.5	Small	Large	200
16	140	4x5	17.5	S7.5	Large	Large	200

Table I-4 140kV Protocols

Section 3.0 N Number Scan Analysis



- 1.) Click the IMAGEWORKS icon (not Exam Rx) to display the images you acquired for the CT N Number Analysis.
- 2.) Use the LIST SELECT function to display the first group of images.
- 3.) Select REVIEW LAYOUTS.

Display the images in groups of four, so you can view all detector rows at once for a single scan. In [Section 2.0](#), you acquired 16 scans (64 images) for each kV station. Select a DFOV of 25cm, and use the default ROI that appears on the screen. When displayed in groups of four, by aperture size:

- n1 (row 2a) = image in the upper left corner; I image location
 - n2 (row 1a) = image in the upper right corner
 - n3 (row 1b) = image in the lower left corner
 - n4 (row 2b) = image in the lower right corner; S image location
- 4.) Select the box ROI. (The box appears over the center of the image; keep this default position.) If you type **prop a** in the Speedbar, the ROI propagates to all 4 images.
 - 5.) Record the CT numbers of water in the corresponding CT N Number Tables:
 - 80kV N numbers: [Table I-5 on page 253](#)
 - 100kV N numbers: [Table I-6 on page 254](#)
 - 120kV N numbers: [Table I-7 on page 254](#)

- 140kV N numbers: [Table I-8 on page 254](#)
 - 6.) If necessary, select SCANNER UTILITIES to open the Utilities window.
 - 7.) Select DETAILED CALIBRATION.
 - 8.) Select CT NUMBER EDITOR.
 - 9.) Select the browser mode.
 - Use the browser to view the old CT numbers for rows 2a, 1a, 1b, and 2b.
 - Select a button on the GUI to display the corresponding techniques

Browser mode deactivates the APPLY button. Leave the browser mode to modify the row 2a, 1a, 1b, and 2b values.
 - 10.) Select UPDATE mode.
- DO NOT engage the Num Lock key on the numeric keypad when you enter the N number values. Remember to check the Num Lock key, and make sure the green light is NOT lit before you start. You can use the numeric keypad to enter the values, regardless of the state of the Num Lock key, but the system refuses to update the N# values with Num Lock engaged. IF you enter the values with Num Lock engaged (green light ON) you will have to reenter all 256 N# values.
- 11.) Check the Num Lock key on the numeric keypad. If necessary, press NUM LOCK to turn OFF the green light on the key cap.
 - 12.) Select a kV to modify.
 - 13.) Type/enter the numbers you recorded in Tables I-5 through I-8 into the editor. Use the arrow buttons to cycle through your entries. Check your entries.
 - 14.) Finish entering the CT N numbers, then select APPLY. Follow this process to enter all kV CT N number adjustments.
- If it has been more than 30 minutes since calibration scans, rerun FASTCAL. N Number Adjustment needs the tube to be warm, or CT numbers will change. FASTCAL takes about 15 minutes.
- 15.) Scan the water phantom and verify the CT number equals 0 ± 1.5 . If the CT number falls outside the 0 ± 1.5 range, repeat [Section 2.0](#) and [3.0](#) until the CT number of the water phantom equals 0 ± 1.5 .

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

80kV	Small Spot Head (Small) Cal 200mA				Small Spot Body (Large) Cal 200mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
80kV	Large Spot Head (Small) Cal 350mA				Large Spot Body (Large) Cal 350mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table I-5 N Number Table for 80kV

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

	Small Spot Head (Small) Cal 200mA				Small Spot Body (Large) Cal 200mA			
100kV	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
	Large Spot Head (Small) Cal 300mA				Large Spot Body (Large) Cal 300mA			
100kV	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table I-6 N Number Table for 100kV

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

	Small Spot Head (Small) Cal 200mA				Small Spot Body (Large) Cal 200mA			
120kV	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
	Large Spot Head (Small) Cal 250mA				Large Spot Body (Large) Cal 250mA			
120kV	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table I-7 N Number Table for 120kV

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

	Small Spot Head (Small) Cal 170mA				Small Spot Body (Large) Cal 170mA			
140kV	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
	Large Spot Head (Small) Cal 200mA				Large Spot Body (Large) Cal 200mA			
140kV	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table I-8 N Number Table for 140kV